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Atomistically-informed cluster dynamics modeling of concurrent interstitial and vacancy a-loop evolution in irradiated alpha-zirconium

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ABSTRACT

A cluster dynamics framework is described that incorporates an extensive database of atomistic simulation results to minimize assumptions in microstructure evolution modeling in irradiated alpha-zirconium. This database informs mechanisms of: 1) defect production behavior in displacement cascades, 2) thermal dissociation rates for all defect clusters, 3) point defect and defect cluster mobility (including anisotropy), 4) spontaneous and thermal drift defect capture at dislocation loops, and 5) cascade overlap effects on defect generation rates. The proposed model mechanistically predicts the concurrent nucleation and growth of interstitial and vacancy a-loops; vacancy loop nucleation is assisted by in-cascade vacancy clustering while vacancy loop growth is stabilized by the incorporation of thermal drift capture radii. Cascade overlap effects on defect generation rates are found to be a necessary component for model predictions of a-loop densities and sizes consistent with the experimental literature. Key characteristics such as average loop diameters, number densities, and the relative ratio of loops with interstitial and vacancy character are in good agreement with experimental observations for neutron-irradiated pure, single-crystalline α -zirconium. As such, the scale-bridging approach between atomistic simulations and cluster dynamics modeling provides several improvements over the assumptions that are commonly implemented in cluster dynamics models and set the stage for future experimental validation.

1. Introduction

One of the issues concerning zirconium cladding performance in nuclear applications is irradiation growth, a volume conservative deformation that occurs in response to an incident irradiation flux [1,2]. This phenomenon is a direct consequence of the anisotropic hexagonal close packed (hcp) crystal structure of α -zirconium, the texture from cladding manufacturing and the unique microstructure that forms under irradiation. A fascinating microstructural feature of irradiated α -zirconium is the co-existence of interstitial and vacancy dislocation loops [3–9]. At low-to-moderate fluence, interstitial and vacancy loops with {1010} prismatic type I (PPI) habit planes (termed a-loops) dominate the microstructure; and following an incubation dose, faulted vacancy loops begin to form on basal planes (termed c-loops) in correlation with the onset of breakaway irradiation growth [4,10–15]. Understanding how α -zirconium microstructure evolves with increasing dose is necessary to improve predictive models of irradiation growth.

Several models have been proposed to explain irradiation growth behavior based on a range of underlying physics, including elastic

interaction differences (EID) between interstitial and vacancy absorption at dislocation loops [16], differences in the anisotropy of diffusion of point defects (DAD) [17–19], and a production bias of mobile defects in displacement cascades [20–23]. Many of these concepts have been included in recent cluster dynamics (CD) models to correlate dislocation loop microstructure evolution with irradiation growth strains. Christien and Barbu developed the first CD model for α -zirconium with an emphasis on the anisotropic diffusion of self-interstitial atoms (SIA) [24]. That model was later modified to include c-loop nucleation at iron clusters to describe breakaway growth behavior [25]. More recently, Li and Ghoniem developed a reduced set of CD equations that considers the mobility of defect clusters [26].

While these models (in addition to those proposed by Golubov et al. [22], Barashev et al. [23], and Choi et al. [27]) have reported success in reproducing experimental irradiation growth strains, there are several inherent assumptions that are not fully consistent with defect production and transport in irradiated zirconium. These assumptions include: 1) only point defects are produced at the NRT standard rate [25,27]; 2) defect clusters do not dissociate [22,23,27]; 3) defect clusters are

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immobile [22,23,25,27,28]; 4) manually-adjusted values for point defect diffusivities are implemented [25–27]; 5) c-loop densities are manually adjusted or saturated rather than mechanistically predicted [22,23,25–27]; and 6) vacancy a-loops either do not grow or are entirely ignored [25–28]. The concurrent growth of interstitial and vacancy a-loops has been a particular challenge in models of microstructure evolution in zirconium. This is a sizeable shortcoming considering that vacancy a-loops can account for over half of the visible microstructure in irradiated zirconium [4,6–9,29]. The model proposed by Christien and Barbu is capable of predicting vacancy a-loop growth, but only for the case of strong SIA anisotropy and preferential thin foil orientation [24]. Likewise, the models of Golubov and Barashev et al. are capable of predicting vacancy a-loop growth, but their model is dependent on a non-isotropic distribution of a-type edge dislocation density [22,23]. Further, within each of these models, the vacancy a-loop number densities are not consistent with experimental observations. The co-existence of interstitial a-loops, vacancy a-loops, and vacancy c-loops has been successfully modeled with an object kinetic Monte Carlo (OKMC) model [30], although the stability of vacancy a-loops is strongly dependent on the parameterization of the model. For many choices of parameters, vacancy a-loops are either unstable, experience limited growth, or experience excessive growth within the OKMC model [30].

Recent atomistic simulations have provided a wealth of knowledge concerning defect production, transport, and interaction with microstructural features. For example, models often assume that only Frenkel pairs are produced from irradiation damage at the NRT standard rate. When considering neutron irradiation, significant in-cascade recombination and clustering is known to occur [31–34]. Production rates of point defects and defect clusters (which directly incorporate recombination and clustering effects) have been recently reported [33,34], and will certainly improve generation rates in higher-order codes. Another aspect lacking in most current models is the treatment of defect diffusivity. Defect clusters are usually considered immobile, monovacancies are generally considered to migrate isotropically, and the diffusivities of point defects are often used as a fitting parameter to provide better agreement with experimental data. Atomistic simulations have instead highlighted strong anisotropic diffusion of both SIAs and monovacancies [35–37] as well as notable mobility of defect clusters [37–39]. Small SIA clusters are reported to take the form of triangular basal platelets which migrate predominantly within the basal plane [37,39] while vacancy clusters exhibit either quasi-isotropic migration or a preference for c-axis diffusion depending on their configuration [37]. Diffusivity data for the a-axis and c-axis migration of point defects and their clusters have been reported for the direct incorporation into mesoscale or continuum models [37]. Finally, the capture radii of point defects at interstitial a-loops, vacancy a-loops, and vacancy c-loops have been reported for spontaneous and thermal drift capture in an effort to improve interaction rate calculations [40]. In this context, the spontaneous capture radius is defined as the distance from the dislocation loop at which the point defect binding energy reaches half of its maximum value. The thermal drift capture radius is defined as the distance from the loop at which the binding energy falls below $3/2k_bT$; at this distance, the defect can no longer thermally escape from the sink interaction gradient. SIA capture is strongly biased to all dislocation loops when considering spontaneous capture; conversely, the thermal drift capture of point defects is sensitive to local stress states near the loops and may encourage vacancy capture at vacancy loops.

The available atomistic modeling database is capable of bridging the considerable length and time scales between molecular dynamics simulations and microstructure evolution. In this manuscript, we describe a cluster dynamics model that is directly informed by this lower-length scale simulation database to minimize assumptions. The model will first be overviewed with an emphasis on how this database has been implemented. Then, the effect of each of these parameters on the nucleation and growth of vacancy and interstitial a-loops in single-crystalline, pure α -zirconium will be highlighted; vacancy c-loops will

be incorporated in a future publication. Finally, a-loop characteristics will be compared with experimentally reported values, and the efficacy of this scale-bridging computational model will be demonstrated.

2. Description of the cluster dynamics model

CD modeling expands upon the classical rate theory approach to consider defect clusters and their interactions [41,42]. The CD approach is at its essence a coupled set of reaction-diffusion equations that governs defect cluster concentrations as a function of time [42]. For this work, only 0-D calculations will be considered such that spatial diffusion and distributions of defect species are ignored. The change in any given cluster population is determined through a balance of gains and losses of that cluster in a given timestep, dt :

$$\frac{dC_i}{dt} = g_i - \bar{D}_i k_i^2 C_i + \sum_{n+m \rightarrow i} k_{n,m}^+ C_n C_m - \sum_{i+j \rightarrow n} k_{ij}^+ C_i C_j + \sum_{n \rightarrow i+j} k_{n-i,i}^- C_n - \sum_{i \rightarrow n+m} k_{ij}^- C_i \quad (1)$$

where C_i is the volumetric concentration of defect species i . The right-hand-side (RHS) terms correspond to production of cluster i from displacement cascades, loss at existing sinks, production of i from cluster interactions, loss of i from cluster interaction, production of i from cluster dissociation, and loss of i from i dissociation, respectively. Each of these terms and the parameters that govern them will be described in the following sections. The key parameters, terms, and their values are tabulated in Table 1 for comprehensive reference.

Table 1

CD model parameters to describe microstructural evolution in neutron-irradiated single-crystalline α -zirconium.

Description	Symbol	Value	Source/Reference
Diffusion within the basal plane	D_i^a	Varies based on Eq. (4)	tables 2 and 3 in [37]
Diffusion along the c-axis	D_i^c	Varies based on Eq. (4)	tables 2 and 3 in [37]
Anisotropy factor of diffusion	p	Varies based on Eq. (6)	[18]
Anisotropic capture efficiency for dislocation lines	$P(\lambda)$	Varies based on Eq. (9)	[18]
Anisotropic capture efficiency for dislocation loops	$P^L(\lambda)$	Varies based on Eq. (13)	[18,46]
Loop ellipticity	e	Vacancy a-loops: 0.80 Interstitial a-loops: 0.95	[8]
Atomic volume	Ω	23.4 Å ³	[47]
Volumetric defect generation rate	g_i	Varies for each cluster, Eq. (2)	[34]
a-type dislocation line density	ρ_d^a	Single crystal: 7.25×10^6 cm ⁻²	[27,48]
c-type dislocation line density	ρ_d^c	Single crystal: 2.25×10^6 cm ⁻²	[27,48]
SIA/dislocation line elastic bias	Z_d^i	1.1	[18,24]
Vacancy/dislocation line elastic bias	Z_d^v	1.0	[18,24]
Spontaneous interaction distance for spherical clusters	r_{ij}^0	6.45×10^{-8} cm	Based on r_{iv} in [24–26]
Spontaneous dislocation loop capture radius	r_d	Varies for each cluster	table 1 in Ref. [40]
Thermal drift dislocation loop capture radius	r_d^h	Varies based on Eq. (21)	table 3 in Ref. [40]
Cluster binding energy	$E_{n,i}^b$	Varies for each cluster, Eq. (26)	[37]
Dpa rate for neutron irradiation	—	8.93×10^{-8} dpa/s	SPECTRA-PKA [44]
Neutron fluence corresponding to 1 dpa of the NRT standard dose	—	1.14×10^{25} n/m ²	[2]

2.1. Defect generation (g_i)

MD simulations have provided insight into the number of defects of each cluster size that may be generated from primary knock-on atom (PKA) cascades of a given energy [31–33,43]. To convert this data into a meaningful parameter for CD modeling, one must consider the expected PKA energy spectrum for a specific neutron flux profile. A representative zirconium PKA spectrum is shown in Fig. 1, as calculated by SPECTRA-PKA [44] for typical PWR operating conditions. The volumetric generation rate for any given cluster i can then be calculated by:

$$g_i = \sum_{E_d=40 \text{ eV}}^{E_{PKA, \max}} N_i(E) * \phi(E) \quad (2)$$

where $N_i(E)$ is the average number of cluster i produced from a PKA of energy E and $\phi(E)$ is the number of PKAs produced with energy E per unit time per unit volume. The total volumetric production of i is determined by summing along the spectrum of possible PKA energies. The energy range is considered from 40 eV, the recommended threshold displacement energy in α -zirconium [45], up to the maximum PKA energy in the estimated spectrum, $E_{PKA, \max}$. $N_i(E)$ has been reported for PKA energies between 1 and 40 keV in Tables 4 and 5 of Ref. [34]. The defect production for PKAs with greater than 40 keV can be extrapolated based on lower PKA energy data. Cascades produced by PKAs with greater than ~ 10 –20 keV are characterized by subcascade formation [33] and the defect production from these events is approximately equivalent to the production from several lower-energy cascades. Using Eq. (2) to sum up the contributions from all PKA energies along the energy spectrum, the calculated volumetric generation rates are plotted in Fig. 2. For these calculations, we considered the tabulated data that included electronic stopping effects, which tend to account for a 10–20 % reduction in the total produced defects [34]. The volumetric rates reported in Fig. 2 for use in Eq. (1) directly account for in-cascade defect recombination and clustering. The dpa rate calculated by SPECTRA PKA is 8.93×10^{-8} dpa/s, and this value is used to convert simulation time into dpa in the figures presented in this paper. The relationship between neutron fluence and dpa is based on the recommended value of Carpenter et al. [2]:

$$1.14 \times 10^{25} n/m^2 (E > 1 \text{ MeV}) = 1 \text{ NRT dpa} \quad (3)$$

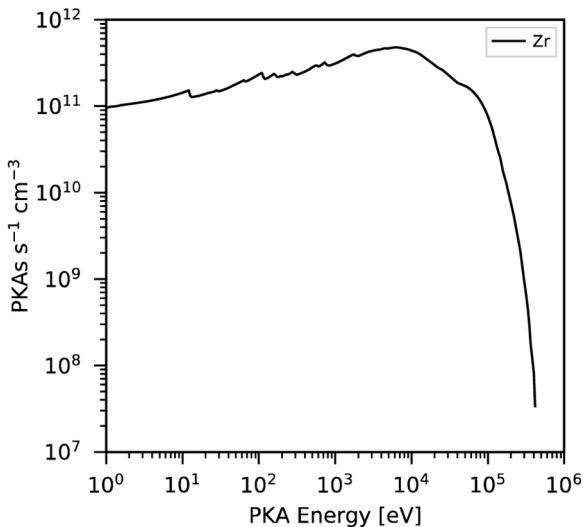


Fig. 1. Estimated zirconium PKA spectrum from a typical PWR neutron flux profile based on calculations performed with SPECTRA-PKA [44].

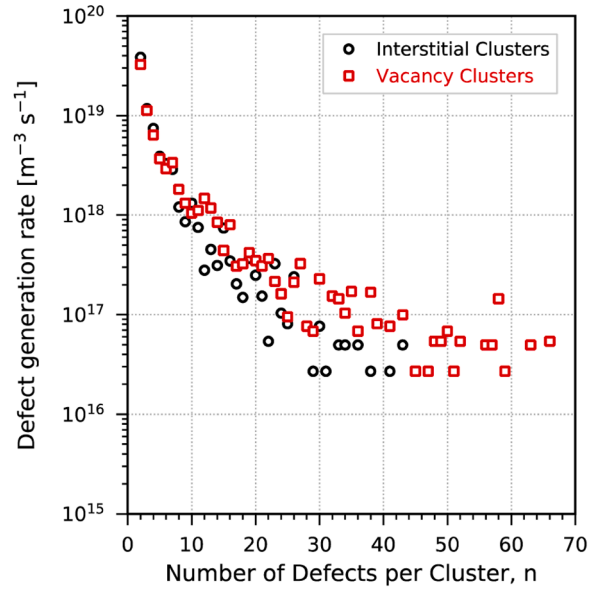


Fig. 2. Volumetric defect generation rates, g_i , as a function of cluster size, as calculated with Eq. (2), in which the red squares represent vacancy and vacancy clusters, and black circles are interstitial and interstitial clusters. These generation rates are based on MD displacement cascades in pristine zirconium [34]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

2.2. Defect diffusivity (D_i) and the anisotropy factor

Interstitial clusters, denoted as “ I_n ” for a size of n , are considered mobile in the size range of $1 < n < 3$. Small interstitial clusters have basal-plane configurations and are highly anisotropic, sluggish diffusers [37,39]. Interstitial clusters with $n > 3$ are characterized by significant anisotropy factors and are poorly described by the current dislocation capture efficiency functions (described in Section 2.3); considering the low mobility of these clusters [37], these clusters are currently considered immobile in this model. Vacancy clusters, denoted as “ V_n ”, are considered mobile in the size range of $1 < n < 6$. Small vacancy clusters assume 3-dimensional configurations and their diffusion anisotropy varies with configuration [37]. While MD simulations in pure zirconium do indicate that larger defect clusters exhibit rapid 1-D migration, the thermal mobility of dislocation loops have been shown to be lower than expected in irradiated bcc Fe [49]. Incorporating this behavior in a CD framework may require a trap-mediated model to account for ballistic de-trapping events [50]; in this work, these clusters are considered immobile, although future models could expand the framework to include such behavior. For each of the mobile species, we consider their diffusion anisotropy. This is done by calculating the diffusivity within the basal plane (D_i^c) and along the c -axis (D_i^a) of each cluster with Eq. (4) using the diffusion pre-factors, D_0 , and migration energies, E_m , from Tables 2 and 3 of Ref. [37]. Based on these two diffusivity values, a mean diffusivity, $\bar{D}_i$, is calculated along with the anisotropy factor of diffusion, p :

$$D_i(T) = D_0 \exp \left(\frac{-E_m}{k_b T} \right) \quad (4)$$

$$\bar{D}_i = (D_i^{a2} D_i^c)^{\frac{1}{3}} \quad (5)$$

$$p = \left(\frac{D_i^c}{D_i^a} \right)^{\frac{1}{6}} \quad (6)$$

I_n and V_n clusters with $n < 9$ are assumed sufficiently small to calculate their radii through a spherical approximation using Eq. (7),

Table 2Summary of experimental data of a-loop characteristics in neutron-irradiated α -zirconium.

	Temperature (K)	Neutron fluence (n/m ²)	Loop Density (m ⁻³)	Mean Diameter (nm)	Interstitial Loops (%)	Ref.
Pure Zr	573	3.8×10^{23}	1.2×10^{22}	9	–	[8]
		1.3×10^{25}	5.0×10^{21}	18.6	–	[61]
	623	2.0×10^{23}	8.0×10^{20}	<10	–	[8]
		5.3×10^{23}	2.7×10^{21}	23	47 – 63	[8]
	668	3.3×10^{23}	7.0×10^{20}	33	23 – 33	[8]
		6.4×10^{23}	2.0×10^{20}	65	9 – 33	[8]
		18.0×10^{23}	2.0×10^{20}	>62	37 – 55	[8]
		3.3×10^{23}	7.0×10^{20}	33	–	[62]
	673	1.0×10^{25}	8.5×10^{20}	54	32	[9]
		1.2×10^{25}	5.0×10^{20}	50	53	[61]
Zircaloy-2	723	1.3×10^{25}	1.0×10^{18}	–	–	[10]
	523	3.1×10^{24}	2.9×10^{22}	8.6	–	[6]
	553 – 603	1.0×10^{25}	1.5×10^{21}	5	–	[29]
	573	1.3×10^{25}	1.9×10^{22}	6.5	–	[61]
	615	1.0×10^{25}	2.6×10^{22}	8.7	–	[6]
	673	1.2×10^{25}	2.8×10^{20}	60	–	[61]
		1.0×10^{25}	6.2×10^{21}	20.1	33 – 45	[6]
Zr-2.5Nb	573	1.3×10^{25}	–	7.0	–	[61]
	673	1.2×10^{25}	–	37	–	[61]

Table 3

Mobile defect diffusivities and anisotropy ratios at $T = 573$ K calculated based on MD data gathered with the BMD19 potential (reported in Tables 2 and 3 in Ref. [37]). The anisotropic capture efficiency for defect capture at interstitial a-loops, $P^{a,I}(\lambda)$, and vacancy a-loops, $P^{a,V}(\lambda)$, is also listed.

Mobile Species	D^a (cm ² s ⁻¹)	D^c (cm ² s ⁻¹)	$\bar{D}$ (cm ² s ⁻¹)	p	$P^{a,I}(\lambda)$	$P^{a,V}(\lambda)$
I_1	7.8×10^{-5}	1.6×10^{-5}	4.6×10^{-5}	0.77	1.3	1.4
I_2	3.1×10^{-5}	4.8×10^{-7}	7.7×10^{-6}	0.50	2.7	3.0
I_3	2.5×10^{-6}	4.8×10^{-9}	3.1×10^{-7}	0.35	5.4	6.1
V_1	1.1×10^{-7}	3.0×10^{-8}	7.0×10^{-8}	0.80	1.2	1.3
V_2	2.2×10^{-7}	5.9×10^{-7}	3.1×10^{-7}	1.17	0.95	0.90
V_3	2.8×10^{-9}	2.0×10^{-9}	2.5×10^{-9}	0.94	1.1	1.1
V_4	9.9×10^{-6}	3.7×10^{-8}	1.5×10^{-6}	0.39	4.3	4.8
V_5	1.6×10^{-8}	1.6×10^{-5}	1.6×10^{-7}	3.18	2.1	1.7
V_6	2.1×10^{-8}	2.4×10^{-6}	2.2×10^{-6}	1.03	1.0	0.98

where Ω is the atomic volume, $23.4 \text{ \AA}^3$. Interstitial clusters I_n and V_n clusters with $n \geq 9$ are considered to be a-loops with PPI habit planes; their radii are calculated based on their disk-like shape through Eq. (8) [47].

$$r_n = \left(\frac{3n\Omega}{4\pi} \right)^{\frac{1}{3}} \quad (7)$$

$$r_n^{PPI} = \sqrt{\frac{acn}{2\pi}} \quad (8)$$

2.3. Defect absorption at pre-existing dislocation line sinks and dislocation loops

For inherently anisotropic materials such as α -zirconium, mobile defects do not migrate equally in all directions and the concept of an unbiased sink is no longer applicable. As such, the sink capture efficiency depends upon the geometry of the sink; sinks with capture volumes or surfaces parallel to the c-axis will preferentially capture basal-plane diffusing defect species and vice-versa [18]. The capture efficiency, $P(\lambda)$, itself will depend upon the angle between the sink surface (or line direction) and the c-axis, λ , as well as the anisotropy factor of the mobile defect diffusion, p [18]:

$$P(\lambda) = \frac{\sqrt{\cos^2 \lambda + p^6 \sin^2 \lambda}}{p^2} \quad (9)$$

Two types of dislocation lines are considered: 1) a-type lines that lie

on PPI planes ($\lambda = 0^\circ$) and 2) c-type lines that lie on basal planes ($\lambda = 90^\circ$). The total sink strength for absorption of mobile cluster i , k_i^2 , is a summation of the sink strengths for these two dislocation types (k_i^{a2} and k_i^{c2} , respectively). The dislocation line densities, ρ_d , for pre-existing a-type and c-type dislocation lines are based on single crystal zirconium estimates: $7.25 \times 10^6 \text{ cm}^{-2}$ and $2.25 \times 10^6 \text{ cm}^{-2}$, respectively [48]. The elastic bias for absorption, Z_i , is assumed to be 1.1 for interstitials and 1.0 for vacancies [18,24,25]. Each type of dislocation line is assigned an anisotropic capture efficiency for absorption, $P(\lambda)$, for the sink strength calculation:

$$k_i^2 = k_i^{a2} + k_i^{c2} \quad (10)$$

$$k_i^{a2} = Z_i \rho_d^a \cdot P(\lambda = 0^\circ) = Z_i \rho_d^a \cdot p^{-2} \quad (11)$$

$$k_i^{c2} = Z_i \rho_d^c \cdot P(\lambda = 90^\circ) = Z_i \rho_d^c \cdot p \quad (12)$$

The calculation of the capture efficiencies of dislocation loops are more complex than for dislocation lines; the angle between dislocation loop line segments and the c-axis vary along the perimeter of the loop. The CD model of Christien and Barbu approximated dislocation loops as hexagons with an equivalent perimeter [24], but this approach has recently been reformulated for a general case. P. Saidi et al. consider a dislocation loop to consist of dislocation lines on a closed path ζ for which the overall capture efficiency is a summation of the contribution from each length segment [46]:

$$P^L(\lambda) = \frac{\int P(\lambda) d\zeta}{\int d\zeta} = \frac{p^{-2} \int_0^{\pi/2} (1 - f(\Theta) + p^6 f(\Theta))^{\frac{1}{2}} (1 - e^2 \sin^2 \Theta)^{\frac{1}{2}} d\Theta}{a \int_0^{\frac{\pi}{2}} (1 - e^2 \sin^2 \Theta)^{\frac{1}{2}} d\Theta} \quad (13)$$

$$f(\Theta) = \left(\left(\frac{b}{a} \right)^4 \left(\frac{1}{\sin^2 \Theta} - 1 \right) + 1 \right)^{-1} \quad (14)$$

$$\cot \lambda = \left(\frac{b}{a} \right)^2 \cot \Theta \quad (15)$$

where a is the semi-major axis, b is the semi-minor axis, and e is the eccentricity of an elliptical loop, respectively [46]. The capture efficiency for prismatic a-loops must be determined through numerical integration of Eq. (13). For basal c-loops, all dislocation line segments are perpendicular to the c-axis, and $P(\lambda)$ is simply equal to the anisotropy factor, p , as calculated with Eq. (9) using $\lambda = 90^\circ$. Interstitial and vacancy a-loops are considered to have semi-axis ratios of approximately

$b/a = 0.95$ and $b/a = 0.80$, respectively [8]. The effect that this has on capture efficiencies is demonstrated graphically in Fig. 3, which compares the capture efficiencies for interstitial a-loops, vacancy a-loops, vacancy c-loops, basal dislocation lines, and prismatic dislocation lines as a function of the mobile defect diffusional anisotropy factor, p . Elliptical a-loops are more likely to capture basal-plane diffusers than circular a-loops due to the fact that a greater fraction of the peripheral line segments have small λ angles.

2.4. Clustering reactions (k_{ij}^+)

There are three broad categories of reactions between clusters i and j : 1) reactions that result in cluster growth ($I_i + I_j \rightarrow I_{i+j}$; $V_i + V_j \rightarrow V_{i+j}$); 2) reactions that result in cluster shrinkage ($I_i + V_j \rightarrow I_{i-j}$, $V_i + I_j \rightarrow V_{i-j}$, where $i > j$); and 3) reactions that result in cluster annihilation ($I_i + V_j \rightarrow \emptyset$, where $i = j$). For clusters with size $n < 9$, the reaction rate constant is based on a spherical reaction volume defined by the interaction distance, r_{ij} [42,51]:

$$k_{ij}^+ = 4\pi r_{ij} (\bar{D}_i + \bar{D}_j) \quad (16)$$

$$r_{ij} = r_i + r_j + r_{ij}^0 \quad (17)$$

The value of r_{ij}^0 is set equal to 6.45×10^{-10} m such that the value of the point defect recombination radius, r_{iv} , equals the commonly implemented value of 1.0×10^{-9} m [24–27]. Large dislocation loops have toroidal reaction volumes due to their disk-like nature and capture tendencies [52]. However, this is only true once the loop radius significantly exceeds the loop defect capture radius, r_d . If this condition is not met, the dislocation loop will either have a spherical or mixed reaction volume. A transition parameter, α_{ij} , allows a smooth transition between spherical and toroidal interaction rates [51,53]. The toroidal contribution to the rate constant is multiplied by the anisotropic capture efficiency of the loop to account for the loop geometry. For a loop i interacting with a mobile defect j [51,53]:

$$k_{ij}^+, \text{ loops} = \left((1 - \alpha_{ij}) z_{ij}^T \cdot P^L(\lambda) + \alpha_{ij} z_{ij}^S \right) (\bar{D}_j) \quad (18)$$

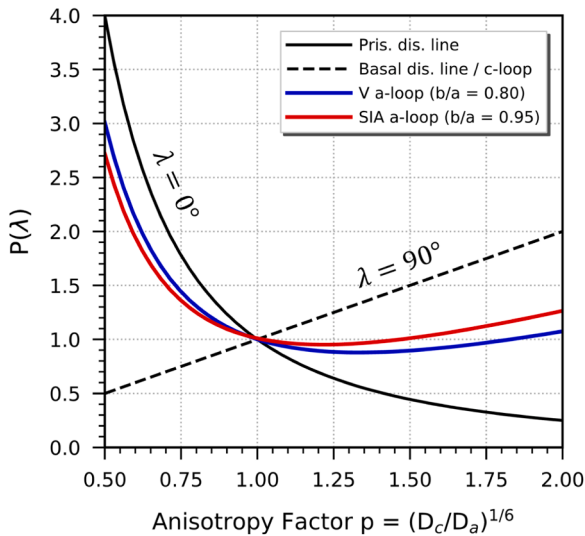


Fig. 3. Capture efficiencies for SIA a-loops (solid red line), vacancy a-loops (solid blue line), prismatic $\langle a \rangle$ -type dislocation lines (solid black line), and basal $\langle c \rangle$ -type dislocation lines and c-loops (dashed black line) are plotted as a function of the mobile defect anisotropy factor, p . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

$$\alpha_{ij} = \left(1 + \left(\frac{r_i}{3(r_j + r_d)} \right)^2 \right)^{-1} \quad (19)$$

$$z_{ij}^S = 4\pi(r_i + r_j + r_d) \quad (20)$$

$$z_{ij}^T = \frac{4\pi^2 r_i}{\ln(1 + 8r_i/(r_j + r_d))} \quad (21)$$

The value of the loop capture radius, r_d , can be based on either spontaneous or thermal drift capture. A spontaneous capture radius is considered the distance at which a defect binding energy to a sink decreases to one half of the maximum value, while thermal drift capture is related to the long-range drift of a defect along an interaction gradient [40]. For spontaneous capture radii, discrete values are available in Table 1 in Ref. [40] and must be interpolated for cluster sizes that are not tabulated. Conversely, spontaneous capture radii are temperature-independent. Conversely, thermal drift radii must be calculated at the selected system temperature using the constants in Ref. [40] with a temperature-dependent expression, where T_m is the melting temperature of zirconium (2128 K):

$$r_d^{\text{thermal}} = \alpha_d \left(\frac{T_m}{T} \right)^{m_d} \quad (22)$$

The capture radii differ based on the size of the defect cluster; the expressions used to calculate size-dependent capture radii for interstitial and vacancy defects are plotted in Fig. 4. Spontaneous radii are smaller than thermal drift radii and favor biased interstitial capture; when considering thermal drift defect capture, dislocation loops tend to favor the capture of the same-type defects (i.e. biased capture of interstitials to interstitial loops and vacancies to vacancy loops) [40].

2.5. Cluster dissociation

In this model, we assume that clusters can only emit monomers of the same type ($I_n \rightarrow I_{n-1} + I_1$; $V_n \rightarrow V_{n-1} + V_1$). All clusters I_n and V_n can thermally dissociate. The reaction rate constant can be calculated as a function of the cluster binding energy and the rate constant for the reverse reaction (e.g. $I_1 + I_{n-1} \rightarrow I_n$; $V_1 + V_{n-1} \rightarrow V_n$) [42]:

$$k_n^- = \frac{k_{1,n-1}^+}{\Omega} \exp \frac{-E_n^b}{k_b T} \quad (23)$$

The binding energy of a monomer i to a cluster of size n can be estimated based on the difference in formation energies between the products and reactants. The formation energies as a function of the cluster size n have been fit to power-law expressions based on MS calculations in Ref. [37]. These expressions capture the energetics of larger dislocation loops well but extrapolate poorly to small defect cluster sizes. As such, separate expressions were fit in this work to the formation energies reported in Ref. [37] for small defect clusters, and are shown in Fig. 5(a). The expressions used in this model for interstitial and vacancy clusters, respectively, are:

$$E_{\text{SIA}}^f = \begin{cases} 2.8n^{0.81}, & 2 \leq n \leq 7 \\ 4.6n^{0.66}, & n > 7 \end{cases} \quad (24)$$

$$E_{\text{V,PPI}}^f = \begin{cases} 1.9n^{0.84}, & 2 \leq n \leq 17 \\ 3.4n^{0.70}, & n > 17 \end{cases} \quad (25)$$

The binding energy is then calculated as:

$$E_n^b = E_1^f - A(n^m - (n-1)^m) \quad (26)$$

where E_1^f is the formation energy of the monomer being emitted (2.94 eV for a single SIA, and 2.03 eV for a monovacancy based on calculations with the BMD19 potential [54]). The size cutoff in Eqs. (24) and (25) were selected to ensure that the binding energies were continuous as a

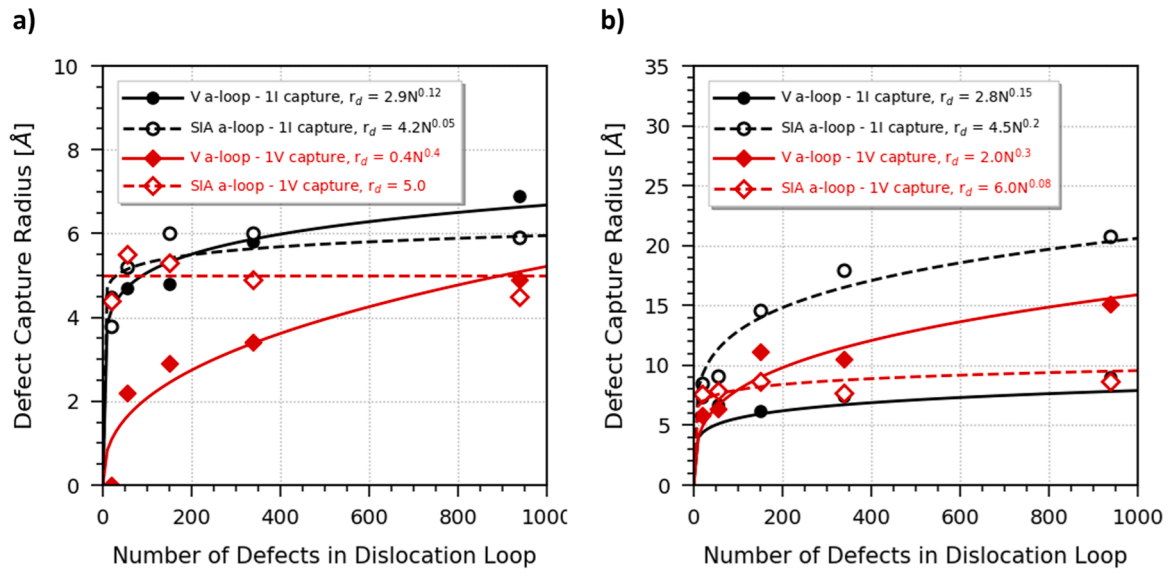


Fig. 4. a) Spontaneous capture radii and b) thermal drift capture radii fits for vacancy (V) and interstitial (SIA) loops. Solid and dashed black lines represent interstitial capture to vacancy and interstitial loops, respectively; solid and dashed red lines represent vacancy capture to vacancy and interstitial loops, respectively. Spontaneous capture data was taken from Table 1 in Ref. [40] (plotted as markers in a)); thermal drift capture data was calculated with Eq. (22) based on parameters from Ref. [40] (plotted as markers in b)). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

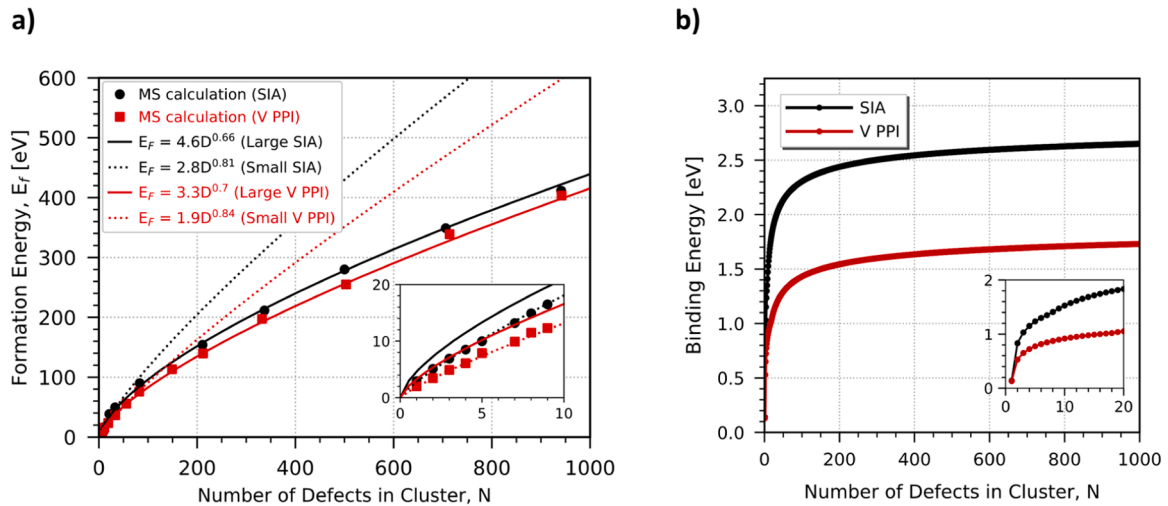


Fig. 5. a) Power-law expressions fit to MS calculations of interstitial (black markers) and vacancy (red markers) cluster formation energies from Ref. [37]. Solid and dashed lines represent fits for small and large defect clusters, respectively. b) corresponding binding energies calculated with Eq. (26) for interstitial (black) and PPI vacancy (red) defects. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

function of size, as plotted in Fig. 5(b).

2.6. Overview of assumptions in the model

A considerable atomistic database is available to inform parameterization of key mechanisms related to defect production, transport, and interaction; these parameters are summarized in Table 1. The primary goal of this work is to directly implement this database in order to reduce assumptions in a physically-relevant CD model. Nevertheless, a few simplifying assumptions are made to reduce the complexity of the model. Each of these points can be improved upon in future work by expanding the framework described here. The five main assumptions are:

- 1) Defects are randomly and uniformly distributed (0-D)
- 2) Dislocation loops are considered immobile

- 3) Only monomers can be emitted during dissociation events
- 4) Dislocation line densities do not evolve with fluence
- 5) Grain boundary sinks are ignored (single-crystal case)

Of these five assumptions, the first is certainly the one which disagrees the most with experiments. Vacancy a-loops are known to align on ordered basal sheets as fluence increases [6–8]. Such spatial ordering may reduce the incoming SIA flux through self-shielding and may play a role in breakaway irradiation growth [29,55]. As for the second assumption, it has been shown experimentally that dislocation loops are far less mobile than MD simulations would imply [37,50,56]. Trap-mediated diffusion may need to be accounted for to include this behavior in the model [50]. Assumption three is valid, as the binding energy of larger defects increases rapidly with size [37]. Assumption four is unlikely to have a large effect considering that irradiation-induced a-loop and c-loop number densities, and

corresponding sink strengths, rapidly exceed the network dislocation density [24,25]. Finally, because we specifically model pure, single-crystalline zirconium, grain boundary sinks are currently ignored and there is no alloying content in this model.

2.7. Cluster dynamics code, Xolotl

The Xolotl cluster dynamics code has been used to model alpha zirconium microstructure evolution. Xolotl was initially developed to simulate gas bubble and defect microstructure evolution in plasma-facing components and structural materials for fusion applications [57–59]. In O-D, a system of coupled ordinary differential equations (ODEs) is generated to track the populations of each defect i as described by Eq. (1). These ODEs are solved using the finite difference method and implicit time integration using the PETSc solver [60]. Each defect of interest is assigned to a species sub-type and characteristic parameters such as the defect radius, diffusivity, anisotropy factor of diffusion, binding energy, and volumetric generation rate are explicitly defined within Xolotl. The possible set of reactions between species sub-types are defined, such as clustering, annihilation, and dissociation. Subsequently, the rates governing all reactions are calculated, and the change in cluster populations within a user-supplied timestep, dt , can be obtained by time integration of the coupled reaction-diffusion ODEs. In this work, we consider only 0-D calculations, although Xolotl is capable of generating a spatial grid to extend calculations to 1-D, 2-D, and 3-D. The timestep is initially set equal to 1 picosecond; each subsequent timestep is increased by a factor of 1.01 to smoothly increase to a maximum timestep of 500 ss.

3. Incremental implementation of physics into the CD model

The CD model proposed here incorporates a wide range of physics that has often been neglected or simplified. For example, cluster dissociation is ignored in many recent models [22,23,27] or is only included for di-interstitials [26]. Despite simulating neutron irradiation, most models only consider point defect generation [22,23,25,27] or only consider small SIA cluster formation, but with unphysically low clustering fractions [26]. Point defect diffusion is often assumed to be isotropic [22,23], or only single SIAs are assumed to migrate anisotropically [25–27]. These models also often use assumed or manually adjusted values for important parameters such as defect diffusivities, clustering fractions in cascades, or loop saturation number densities.

Common assumptions and input parameters that have been implemented in recent models are described and applied to our Xolotl CD model. This provides for computational verification against literature results and the insight associated with systematically including, or not, common assumptions on the resulting Xolotl CD predictions. Once it is demonstrated that these parameters and assumptions do not accurately predict microstructure evolution in irradiated zirconium, the robust dataset detailed in Table 1 is incrementally introduced to the CD model. As such, the effect of each of the physical parameters will be showcased individually. In Section 4.5, the CD model predictions are shown after incorporating each of the parameters informed by Table 1. Finally, a key modification to the defect production rates is outlined in Section 4.6 to account for cascade overlap effects (i.e. when displacement cascades overlap an existing density of defects as opposed to traditional simulations performed in pristine zirconium).

In order to provide a preliminary comparison of Xolotl CD model predictions to experimental results, Table 2 summarizes literature reports for a-loop number densities, mean diameters, and the relative percent of interstitial to vacancy a-loops for a range of temperatures in pure and alloyed zirconium. Quite a few factors simultaneously affect a-loop characteristics. An increase in irradiation temperature leads to increased loop size and decreased density, or loop coarsening. A similar effect is noted for an increase in the neutron fluence at a given temperature. For example, at 573 K, increasing neutron fluence from 3.8×10^{23}

n m^{-2} to $1.3 \times 10^{25} \text{ n m}^{-2}$ at 573 K results in an increase in the average a-loop diameter from 9 nm to 18.6 nm with a corresponding decrease in their number densities in pure zirconium. It is clear from this experimental data that there is a greater fraction of interstitial loops (relative to vacancy loops) at lower temperatures; the reverse is true at higher temperatures. Finally, there is clearly a significant effect of alloying elements, although the modeling results presented in this paper will not address trap-mediated defect diffusion. The data in Table 2 indicates that a-loops form with notably smaller diameters and higher number densities in alloyed zirconium. In this manuscript, we focus on the lower-temperature irradiation data for pure zirconium. In particular, we focus our model predictions on the two data sets at 573K: the low-fluence $3.8 \times 10^{23} \text{ n m}^{-2}$ (0.033 dpa) data reported by Jostsons et al. [8] and the higher-fluence $1.3 \times 10^{25} \text{ n m}^{-2}$ (1.15 dpa) data reported by Gilbert et al. [61]. The a-loop size distribution histograms reported in these studies are re-produced in Fig. 6 for ease of comparison. At 0.033 dpa, we expect a narrow distribution of a-loops with a mean diameter of 9 nm ($1.2 \times 10^{22} \text{ m}^{-3}$ density); at 1.15 dpa, we expect a wider distribution of a-loops with a mean of 18.6 nm ($5 \times 10^{21} \text{ m}^{-3}$ density).

3.1. Xolotl model benchmarking past literature results using commonly assumed inputs

Common assumptions and input values in prior α -zirconium microstructure evolution models include:

1. Only point defects are produced by neutron damage at the NRT standard rate ($G = 10^{-7} \text{ dpa s}^{-1} = 4.3 \times 10^{15} \text{ point defects s}^{-1} \text{ cm}^{-3}$) [24–27].
2. Clusters do not dissociate [22,23,27].
3. Defect clusters are immobile [22–25,27].
4. Single interstitials diffuse with a manually adjusted value of $\bar{D}_i = 10^{-6} \text{ cm}^2 \text{ s}^{-1}$ and an assumed anisotropy factor of $p = 0.765$ [24–27].
5. Monovacancies migrate isotropically with a manually adjusted value of $D_v = 3.0 \times 10^{-17} \text{ cm}^2 \text{ s}^{-1}$ [24–27].

In order to demonstrate how well these assumptions and parameters represent the underlying mechanisms of irradiation damage evolution, we incorporated these values into our Xolotl CD model described in Section 2. The resultant damage evolution is demonstrated in Fig. 7 ($T = 573 \text{ K}$). In agreement with past literature [24–27], the point defect concentrations diverge at low fluence and eventually saturate at 1.0 dpa. The difference in saturated point defect concentrations is driven by the

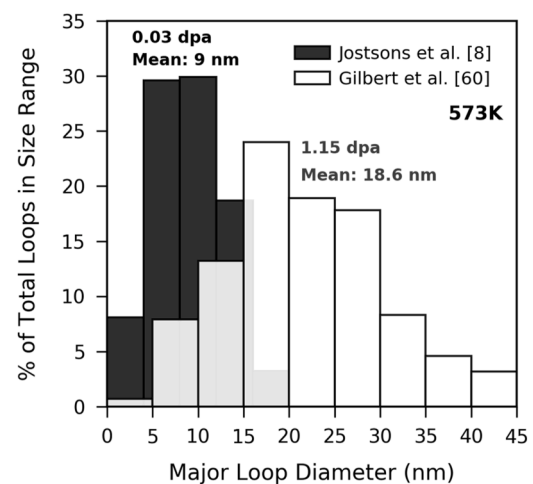


Fig. 6. Dislocation a-loop size distribution histograms reported in pure α -zirconium as reproduced from Jostsons et al. (0.033 dpa, filled grey bars) [8] and Gilbert (1.15 dpa, open bars) [61] at 573K.

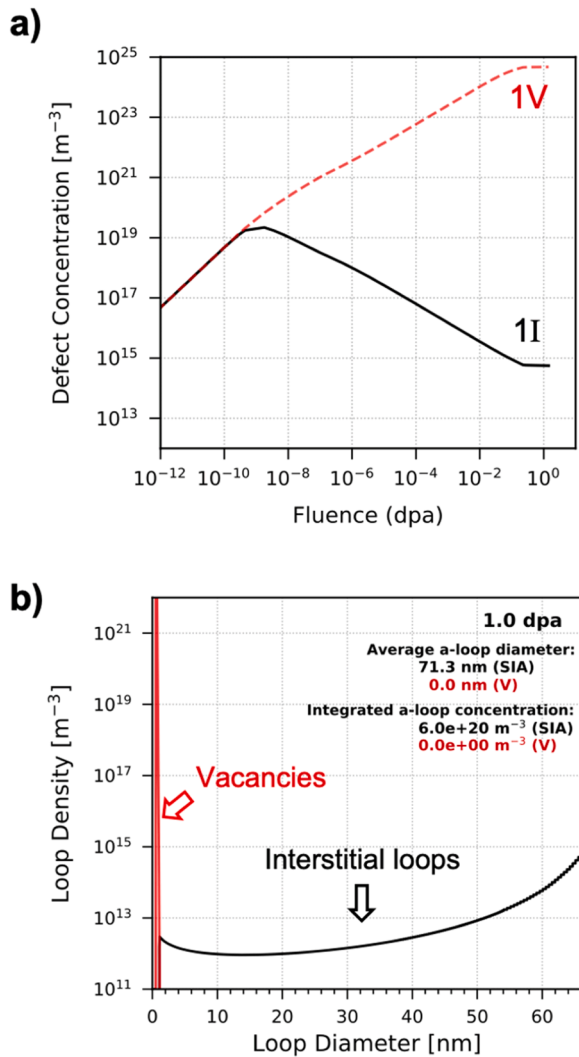


Fig. 7. CD model predictions at $T = 573$ K for a) single vacancy (dotted red line) and single interstitial (solid black line) concentrations as a function of fluence and b) vacancy (red) and interstitial (black) a-loop size distributions at 1.0 dpa when incorporating common assumptions and parameter values from literature. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

considerable discrepancy in their diffusivities ($\bar{D}_i = 10^{-6} \text{ cm}^2 \text{ s}^{-1}$, $D_v = 3.0 \times 10^{-17} \text{ cm}^2 \text{ s}^{-1}$). Monovacancies are largely unable to interact with surrounding sinks until concentrations grow large enough to compensate for low diffusivities. The effect that this has on a-loop evolution with fluence is clear in Fig. 7(b)—vacancy a-loops do not nucleate or grow.

This behavior has been reported with past CD models [25], and due to the difficulty in simultaneously growing vacancy and interstitial a-loops, vacancy a-loops are ignored in most recent modeling studies [26–28]. It is experimentally established that vacancy a-loops form in both pure and alloyed zirconium and are even more abundant than interstitial a-loops for many irradiation conditions [3,4,6–8,11,61]. For this reason, we conclude that these common assumptions and parameter values should not be used in future modeling studies, as they do not accurately produce a physically-relevant damage evolution profile. In the following sections, we will incrementally incorporate the more robust dataset described in Table 1 to demonstrate improvements in the Xoloti CD model predictions.

3.2. Effect of defect cluster generation rates based on MD cascade data (g_i)

We first address the common assumption that only point defects are produced from neutron irradiation at the standard NRT rate. It is widely reported from MD simulations that the surviving fraction of the NRT-predicted number of defects is a relatively small percentage ($\sim 30\%$) [31–34]. Applying defect generation at the NRT standard rate corresponds to a sizeable over-estimation of the defect source term. Furthermore, a significant fraction (up to 40–60 %) of surviving defects are contained in defect clusters. In fact, even large dislocation loops can directly nucleate from high-energy cascade events [33,34]. Therefore, the first improvement that we implement to the CD model is incorporating the volumetric defect generation rates presented in Fig. 2. Because the volumetric generation rates are based on MD cascade data, they inherently incorporate in-cascade recombination and defect clustering, although these volumetric generation rates do not account for the additional correlated recombination that occurs as the mobile species diffuse through and ultimately away from the less mobile cascade debris of a single displacement cascade.

Fig. 8 demonstrates the effect of incorporating the volumetric generation rates of defect production into the model prediction. Vacancy a-loop growth is stabilized while interstitial a-loop growth is reduced. There are two main contributions that explain this behavior: 1) fewer mobile point defects are produced due to in-cascade recombination and defect clustering ($\dot{g}_{1I} = 4.14 \times 10^{14} \text{ cm}^{-3} \text{ s}^{-1}$, $\dot{g}_{1V} = 3.0 \times 10^{14} \text{ cm}^{-3} \text{ s}^{-1}$ compared to the traditional $\dot{g}_{\text{NRT}} = 4.3 \times 10^{15} \text{ cm}^{-3} \text{ s}^{-1}$) and 2) defect clusters up to a size of 70 defects are directly generated within the cascades. These two factors assist nucleation of vacancy a-loops and reduce the growth of interstitial a-loops. The total concentration of mobile SIAs is overall reduced due to the higher density of available clusters to interact with. However, it is clear that this alone is not sufficient to accurately model a-loop evolution, as the average diameters are much smaller than experimentally-reported values presented in Table 2.

3.3. Effect of dissociation rates

The magnitude of the dissociation rate constant is dependent on two factors: 1) the monomer binding energy to the parent cluster and 2) the forward rate constant for the inverse clustering reaction [42]. The former is calculated with Eq. (26) based on molecular statics calculations of defect cluster formation energies (Fig. 6). The forward rate constant is calculated with Eq. (16) assuming either spherical clusters or Eq. (18) for dislocation loop geometries; both of these expressions depend directly on the diffusivity of the monomer being emitted from the cluster. As the diffusivity of the monomer decreases, the rate of dissociation correspondingly decreases; conceptually, a less mobile monomer is more likely to be re-captured by the cluster from which it was emitted. The dissociation rate constants are compared in Fig. 9 for two different cases: 1) using the commonly-assumed values for point defect diffusivities in literature (dashed lines) and 2) using the diffusivities calculated through MD simulations with the BMD19 potential (solid lines).

When considering common literature values for point defect diffusivities, the dissociation rate constants for vacancy clusters are several orders of magnitude lower than equivalent interstitial clusters, despite the lower binding energies of vacancy clusters. Conversely, when performing the calculation with diffusivities derived from MD simulations (tables 2 and 3 in Ref. [37]), vacancy clusters show notably larger dissociation rates than interstitial clusters. This is now consistent with the concept of the production bias theory which relies on the thermal dissociation of vacancy clusters at higher temperatures to describe key irradiation phenomena [20,21]. It is clear that the common monovacancy diffusivity value often used in modeling attempts is too low to

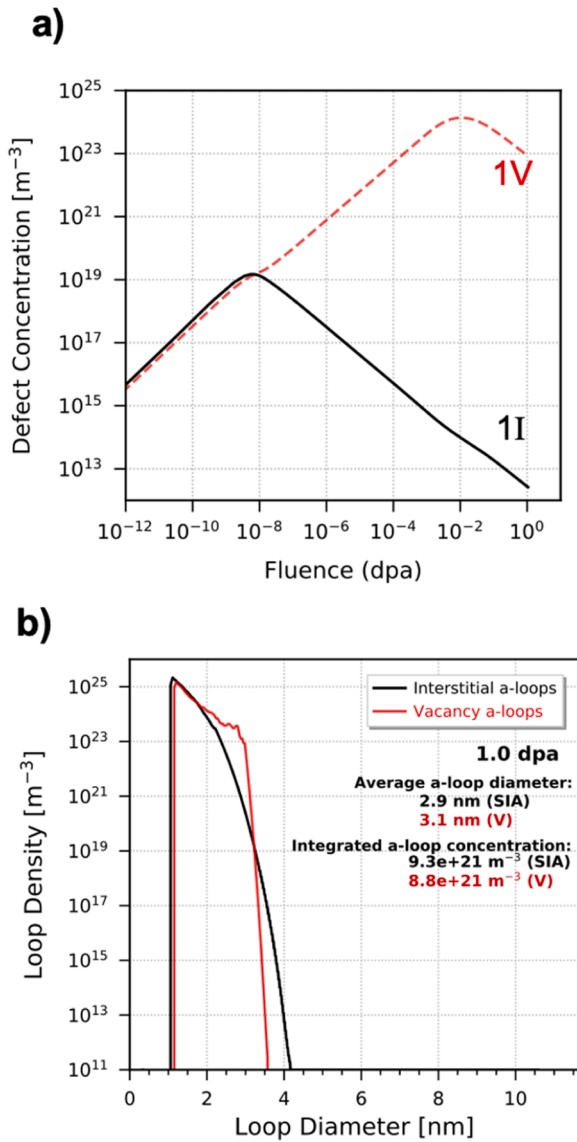


Fig. 8. CD model predictions at $T = 573$ K for a) single vacancy (dotted red line) and single interstitial (solid black line) concentrations as a function of fluence and b) vacancy (red) and interstitial (black) a-loop size distributions at 1.0 dpa when incorporating common assumptions and parameter values from literature. The simulation is modified from Fig. 7 by using the volumetric defect generation rates calculated with MD displacement cascade data (Fig. 2) rather than assuming simple point defect generation at the NRT standard rate. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

properly describe damage evolution behavior. The effect of the updated dissociation rates shown in Fig. 9 will be demonstrated in the following section with the implementation of MD-based point defect and defect cluster mobilities into the CD model.

3.4. Effect of point defect and defect cluster mobility

Defect mobility is incorporated into the CD model based on the parameter datasets calculated directly from MD simulations [37]. The calculated diffusivities, anisotropy factors, and anisotropic capture efficiencies for interstitial and vacancy a-loops at $T = 573$ K are tabulated in Table 3. Both types of point defects migrate with a preference for basal-plane diffusion ($p < 1$). Interstitial clusters are predominantly basal plane diffusers ($p \ll 1$) while vacancy clusters generally migrate

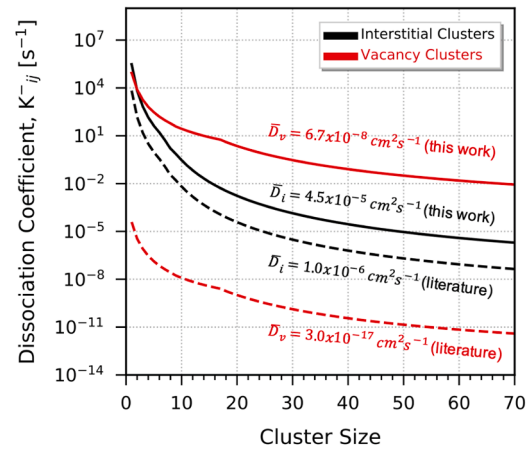


Fig. 9. Dissociation rate coefficients at $T = 573$ K for interstitial clusters (black) and vacancy clusters (red) as a function of cluster size. Solid curves are based on calculations with the defect diffusivities reported in tables 2 and 3 in Ref. [37] while dashed curves consider commonly assumed diffusivities in literature [24–27]. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

quasi-isotropically ($p \sim 1$) or with a preference along the c-axis ($p > 1$) depending on the cluster size [37]. Mobile defects with low anisotropy factors will preferentially interact with prismatic a-loops; the lower the anisotropy factor, the stronger the bias for their capture at a-loops. In addition, vacancy a-loops are more elliptical than interstitial a-loops [8] and this results in an elongation of these loops along the c-axis. As a consequence, mobile defects with low anisotropy factors will have a greater bias for capture at vacancy a-loops than interstitial a-loops, as is seen in Fig. 3 and Table 3. Furthermore, MD simulations [37] and *ab initio* calculations [35] both indicate that vacancy diffusion is more rapid than has been assumed in previous models [24–27]. This is consistent with the “anomalous self-diffusion” and “facile vacancy migration” that has been coined based on Hood’s experimental observations in α -zirconium [63].

We first analyze the effect of cluster dissociation and defect mobility on the concentrations of point defects and small defect clusters in Fig. 10. The most notable differences are for monovacancies and vacancy clusters, for which the average defect concentrations are orders of magnitude lower than predicted using the manually-adjusted diffusivity values from literature. The reduction in monovacancy concentrations is predominantly driven by enhanced interaction with sinks and defect clusters due to the larger diffusivity value predicted by MD simulations ($\bar{D}_v = 7.0 \times 10^{-8} \text{ cm}^2 \text{ s}^{-1}$ compared to $D_v = 3.0 \times 10^{-17} \text{ cm}^2 \text{ s}^{-1}$ at 573 K) which enhances cluster dissociation rates (Fig. 9).

The effect of defect mobility on dislocation loop evolution is highlighted in Fig. 11, which should be compared to Fig. 8. We now observe considerable interstitial loop growth and notable vacancy loop growth; the number densities of both types of loops are in the range predicted by experimental data at 573 K (Table 2). While the average loop diameters have also increased, they are still below the experimental measurements at a dose of 1.15 dpa. For this irradiation condition, Gilbert et al. observed average a-loop diameters of 18.6 nm as seen in Fig. 6 [61]. The magnitude of the cluster reaction rate is dependent on the dislocation loop defect capture radii (which are still considered to be zero in the CD calculation plotted in Fig. 11). In the following section, we demonstrate the effect that spontaneous and thermal drift dislocation loop capture radii have on the predicted loop size distributions.

3.5. Effect of dislocation loop defect capture radii

Dislocation loop capture radii influence the sharpness of the transition between spherical and toroidal clustering kinetics as well as the

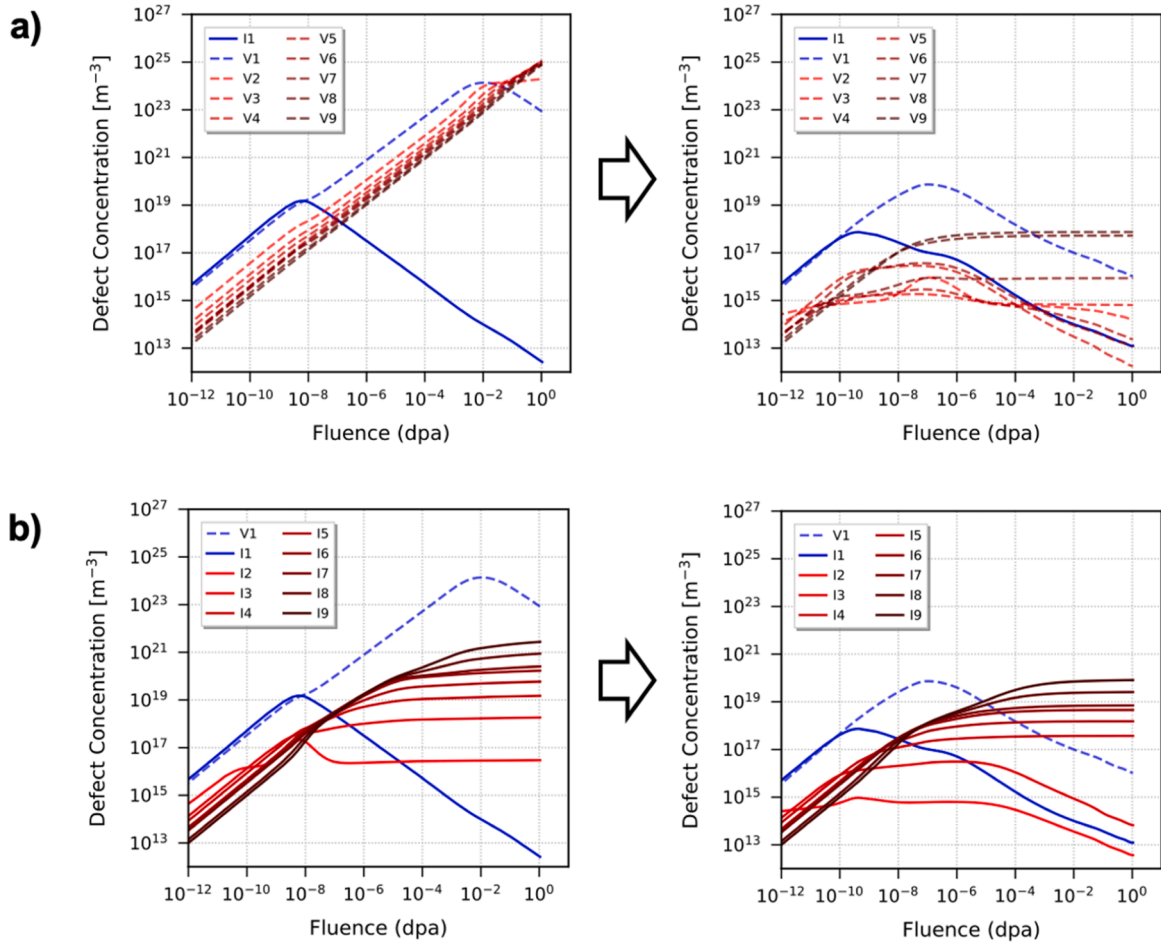


Fig. 10. Defect concentrations for a) vacancy (dashed curves) and b) interstitial (solid curves) defects in the size range of $1 \leq n \leq 9$ is shown before and after incorporating cluster dissociation for all clusters and MD-based diffusivities at $T = 573\text{K}$. The legend defines the specific vacancy or interstitial cluster size.

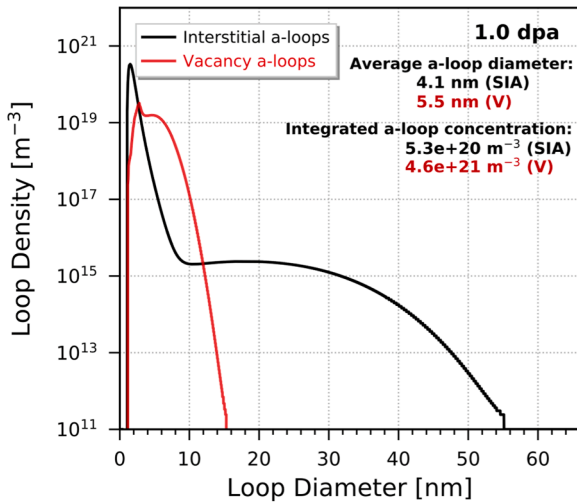


Fig. 11. CD model predictions at 573 K for the vacancy (red) and interstitial (black) a-loop size distributions at 1.0 dpa after incorporating the volumetric defect generation rates, defect mobility, and dissociation rates based on the dataset outlined in Table 1. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

magnitude of the toroidal reaction rate. Considering spontaneous defect capture, most dislocation loops experience biased capture of interstitials [40]; when considering the thermal drift method, an inherent bias

develops for the capture of same-type defects at dislocation loops (i.e. vacancy capture at vacancy loops). The effect of the capture radii on point defect interaction rates with a-loops is demonstrated in Fig. 12. In this figure, the reaction rates are calculated with no capture radii ($r_d = 0$), spontaneous capture radii, and thermal drift capture radii. For the case with no capture radii, capture rates are simply based on the size of the loop as well as the size and mobility of the captured defect. Incorporating spontaneous radii increases the interaction rate magnitude as a result of the larger reaction volume. Because the spontaneous capture radii are similar for all dislocation loops, there is a slight bias for point defect capture at vacancy a-loops due to the greater anisotropic capture efficiency. Conversely, incorporation of thermal drift radii leads to a greater deviation depending on the type of captured defect and the type of loop. Due to the inherent bias for the capture of same-type defects, interaction rates are higher between single SIAs and interstitial a-loops as well as between vacancies and vacancy a-loops (Fig. 12(c)).

The resultant size distribution of dislocation loops using spontaneous and thermal drift capture radii are plotted in Figs. 13(a) and (b), respectively. These simulations were performed with the full dataset outlined in Table 1. When incorporating spontaneous capture radii, the Xolotl model predictions are similar to the case with no capture radii (Fig. 11). While the reaction rates are all larger in magnitude, the SIA bias for absorption at dislocation loops drives interstitial loop growth. The necessity for a mechanistic bias, then, is evident in order to drive simultaneous vacancy loop growth. Incorporating thermal capture radii into the CD model results in the simultaneous growth of interstitial and vacancy a-loops as seen in Fig. 13(b) at 1.0 dpa and $T = 573\text{K}$. However, based on the experimental results of Gilbert et al. [61], the resulting

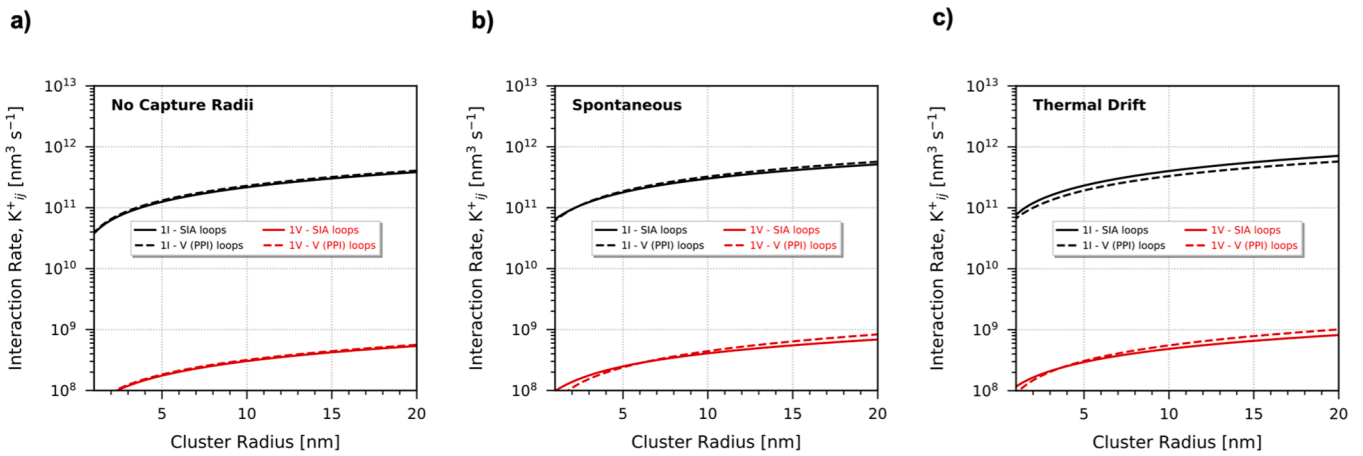


Fig. 12. Point defect interaction rates with dislocation loops as a function of the loop radius when calculated using a) no capture radii, b) spontaneous capture radii, and c) thermal drift capture radii. Black and red curves represent single interstitial and single vacancy capture, respectively; solid and dashed curves represent defect capture at interstitial and vacancy a-loops, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

a-loop number densities and diameters are over-predicted.

3.6. Accounting for defect production saturation due to cascade overlap effects

The volumetric generation rates for defects shown in Fig. 2 are based on MD displacement cascades in a pristine simulation cell free of any pre-existing defects. As such, the defect generation data is only representative of very low doses for which defect concentrations remain dilute. It has been demonstrated recently in pure W that the reduction in thermal diffusivity saturates at a dose of ~ 0.1 dpa under continuous 20 MeV self-ion irradiation at room temperature [64]. When calculating the Frenkel pair density correlated with the reduction in thermal diffusivity, the defect production efficiency deviates from the expected behavior at ~ 0.005 dpa. Consecutive MD displacement cascades were later performed in a single simulation cell of pure W, and defect production was found to saturate below a dose of ~ 0.1 dpa due to cascade overlap effects [65]. In α -zirconium, there is experimental evidence that the defect yield also begins to saturate at a dose on the order of 0.1 dpa [66] and sequential displacement cascade simulations predict defect production saturation in the range of 0.01 to 0.1 dpa [67]. Based on these observations, volumetric generation rates in CD models should not remain constant as defect concentrations increase and should instead begin to be reduced as cascades begin to overlap existing defects.

Ideally, cascade overlap effects would be incorporated into the CD model in such a way that existing defect populations are affected by cascade events. This would require a robust implementation capable of tracking cluster growth or shrinkage, but the data required to adequately inform such an implementation has not yet been reported. As such, we have included cascade overlap effects in a simpler form: namely, with a reduction in the volumetric defect generation rates with increasing dose, d . The “effective” generation rates, $\dot{g}^*(d)$, are defined in terms of a cascade efficiency factor, $\eta(d)$, and the original volumetric generation rates, $\dot{g}$:

$$\dot{g}^*(d) = \dot{g} \cdot \eta(d) \quad (27)$$

$$\eta(d) = [1 - \tanh(C_1 \cdot d - C_2)] * \frac{1 - C_3}{2} + C_3 \quad (28)$$

The constant C_1 controls the steepness of the hyperbolic curve with increasing dose, C_2 controls the onset dose for cascade overlap effects, and C_3 determines the residual defect generation rate once cascade overlap effects have saturated. Three potential hyperbolic tangent fits are plotted in Fig. 14 to demonstrate their effects on microstructural

evolution predictions: η_1 , η_2 , and η_3 which correspond to saturation at 0.01, 0.1, and 1.0 dpa, respectively. The closest agreement with experimental data is found when implementing the η_2 curve; this is fully consistent with the saturation dose estimated in literature for pure W and α -zirconium [64–66]. Approximate values for the C_1 , C_2 , and C_3 constants can be informed by the MD data presented in Fig. 1 of Ref. [67] which plots the number of stable defects with increasing dose from sequential displacement cascades. This data is reproduced in Fig. 15(a) to showcase the difference in the Frenkel pair generation rate at low dose (i.e. pristine zirconium) and at a higher dose once defect populations have increased. The residual defect generation fraction can be estimated based on the slopes of the generation rates in the low-dose and higher dose regions; this analysis provides a C_3 value of 0.0075 (a residual generation rate that is 0.75 % of the original rate). The C_1 and C_2 constants were then selected such that defect generation would saturate within the range of 0.01 to 0.1 dpa as reported in Refs. [64–66]. Values of $C_1 = 47$ and $C_2 = 1.65$ reproduce this behavior well, as can be seen with the corresponding cascade efficiency curve plotted in Fig. 15(b).

With the incorporation of the full dataset in Table 1 and the optimized defect generation rates accounting for cascade overlap, the resulting Xolotl predicted prismatic dislocation loop populations are shown in Figs. 16(a),(b) and (c) at 0.033, 1.15, and 5.0 dpa. Corresponding loop size distribution histograms are shown in Figs. 16(d),(e) and (f), respectively. These histograms are plotted as stacked vertical bars to demonstrate the total combined contribution of vacancy and interstitial a-loops. This was done to enable comparison with the experimental loop size distributions plotted in Fig. 6 which did not distinguish between interstitial and vacancy a-loop contributions. The results at 0.033 dpa can be compared to the work of Jostsons et al. at 573 K for which the average reported a-loop diameter and number density are 9 nm and $1.2 \times 10^{22} \text{ m}^{-3}$, respectively [8]. Similarly, the results at 1.15 dpa can be compared to the work of Gilbert et al. at 573 K, with an average a-loop diameter and number density of 18.6 nm and $5 \times 10^{21} \text{ m}^{-3}$, respectively [61]. The experimental size distributions from Fig. 6 reported by Jostsons et al. and Gilbert et al. are plotted in Fig. 16 (d) and Fig. 16(e), respectively, to facilitate comparisons between the CD model predictions, which is in good agreement with these reported a-loop characteristics. While the relative fraction of interstitial-to-vacancy a-loops is not typically reported at 573 K, the general trend from higher-temperature irradiations (Table 2) suggests that a near-equal ratio should be expected, as is predicted by our CD model. The model considers a-loop coarsening with increasing irradiation dose, resulting in a decrease in predicted number densities (Fig. 17 (a)) and a corresponding increase in the average loop diameters (Fig. 17

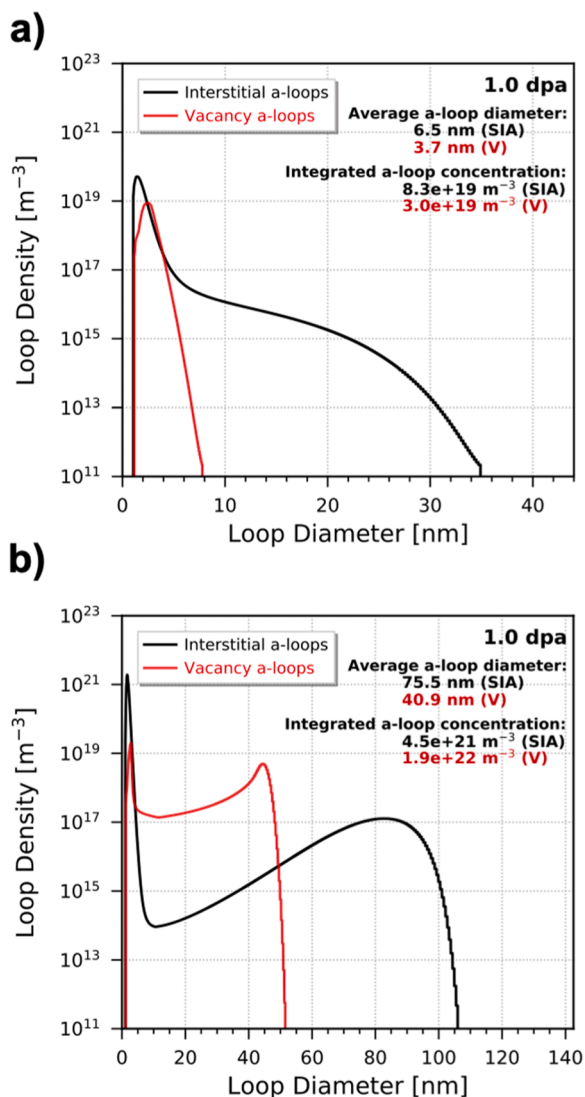


Fig. 13. CD model predictions at 573 K for the vacancy (red) and interstitial (black) a-loop size distributions at 1.0 dpa using the full dataset and framework described in this work. These simulations were run using a) spontaneous capture radii and b) thermal drift capture radii, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(b)). At higher doses, a-loop number densities saturate while average diameters continue to grow. Estimates of average loop size and density consider only loops with diameters greater than 3 nm to account for an approximate resolution limit with TEM. This behavior is predominantly driven by the C_3 constant in the cascade efficiency factor (Eq. (28)) which controls the residual defect generation rate once cascade overlap effects have saturated. For lower values of C_3 , the average a-loop diameter grows more slowly with increasing dose.

The low-dose saturation of irradiation growth strains in irradiated zirconium has been attributed to differences in the growth rates of interstitial and vacancy a-loops [68]. This concept has been investigated through CD model predictions of interstitial and vacancy a-loop number densities with increasing dose in Fig. 18. Interstitial a-loop densities are initially lower than vacancy a-loops at very low doses (~ 0.0001 dpa) but quickly outnumber vacancy a-loops at ~ 0.002 dpa. This behavior in the model is related to differences in loop nucleation rates from displacement cascades. Vacancy a-loops are capable of nucleating directly from displacement cascades due to the large size of vacancy clusters that form at the vacancy-rich cascade core. This is much less likely to occur for

interstitial a-loops, as interstitials usually form smaller defect clusters near the cascade periphery. Interstitial clusters predominantly coarsen into interstitial a-loops through interstitial capture over time. With additional coarsening, interstitial a-loop number densities decrease with a corresponding increase in average loop diameter (Fig. 17(b)). The difference in a-loop nucleation rates directly from displacement cascades and the time needed for interstitial clusters to coarsen into large a-loops may both be contributing factors to the delayed saturation of irradiation growth strains at lower doses.

It is evident from this work that the defect production rate predicted by MD displacement cascades in pristine zirconium (Fig. 2) results in a significant overprediction of microstructure growth under irradiation. We believe this is due to the fact that the initial production rate is greatly reduced through a combination of factors, including point defect recombination that occurs during diffusional timescales [69] and the defect reactions that occur once extended defects build up in the material (as is supported experimentally in Ref. [66] and through MD simulations in Ref. [67]). This is implemented as a reduction in the cascade efficiency factor with increasing dose (Eq. (27) and Fig. 15(b)). Realistically, the cascade efficiency factor is likely also compensating for other effects which inhibit defect production, such as correlated recombination [69], that are not currently included in the model. Incorporating mechanisms which reduce defect production rates compared to traditional MD cascade predictions is considered as a necessary component to adequately model dislocation loop microstructure evolution. Without these effects, defect growth is erroneously rapid (for example, compare Fig. 13(b) with Fig. 16(b) for defect size distributions without and with cascade overlap effects, respectively). Future work will continue to elucidate mechanisms affecting defect production and should quantify these effects in a manner which can be directly incorporated in CD modeling.

The implementation of the cascade efficiency factor chosen in Eq. (28) alters the defect generation rate as a function of dose. However, overlap effects should realistically depend on the integrated defect density or, alternatively, on the total defect volume to physically represent the underlying mechanisms. For a given temperature and irradiation condition, dose can act as a surrogate for these properties, but the chosen efficiency curve in Fig. 15(b) may not be applicable for other irradiation conditions as we expect this to depend on irradiation temperature and dose rate. In order to inform a generalized efficiency curve, the CD model at $T = 573$ K with the efficiency curve selected in Fig. 15(b) was used to calculate the predicted Frenkel pair (FP) density at a range of doses between 0.0001 and 10 dpa. The integrated FP density at a given dose was calculated by summing up the total number of interstitials in the system. The cascade efficiency originally plotted in Fig. 15(b) was then plotted against the integrated FP density rather than the corresponding dose in Fig. 19. This analysis clearly demonstrates that there is a critical defect density (approximately 10^{24} FP/m³) at which cascade overlap effects are expected to initiate. The cascade efficiency factor decreases sharply until it saturates at a defect density of approximately 10^{25} FP/m³. This behavior can be reproduced with an equation of an equivalent form to Eq. (28) using the constants $C_1 = 5.3 \times 10^{-25}$, $C_2 = 3.7 \times 10^{24}$, and $C_3 = 0.0075$ for FP densities in units of m⁻³. Correlating the reduced efficiency factor with defect density allows greater transferability of the model to other irradiation conditions. It is recommended that a similar efficiency factor be implemented in future CD models for zirconium and other material systems.

The resulting CD framework and complete model (e.g., predictions demonstrated in Figs. 16 and 17) described in this work agrees well with the selected experimental data for modeling microstructure evolution in neutron-irradiated pure zirconium. There are, however, three areas that require future work: 1) basal c-loop nucleation and growth, 2) the effect of alloying elements and impurities, and 3) spatial dependence, including possible extension of the model to account for elastic interactions, of the defect cluster populations. The current CD model can naturally be extended to incorporate basal c-loop nucleation and growth

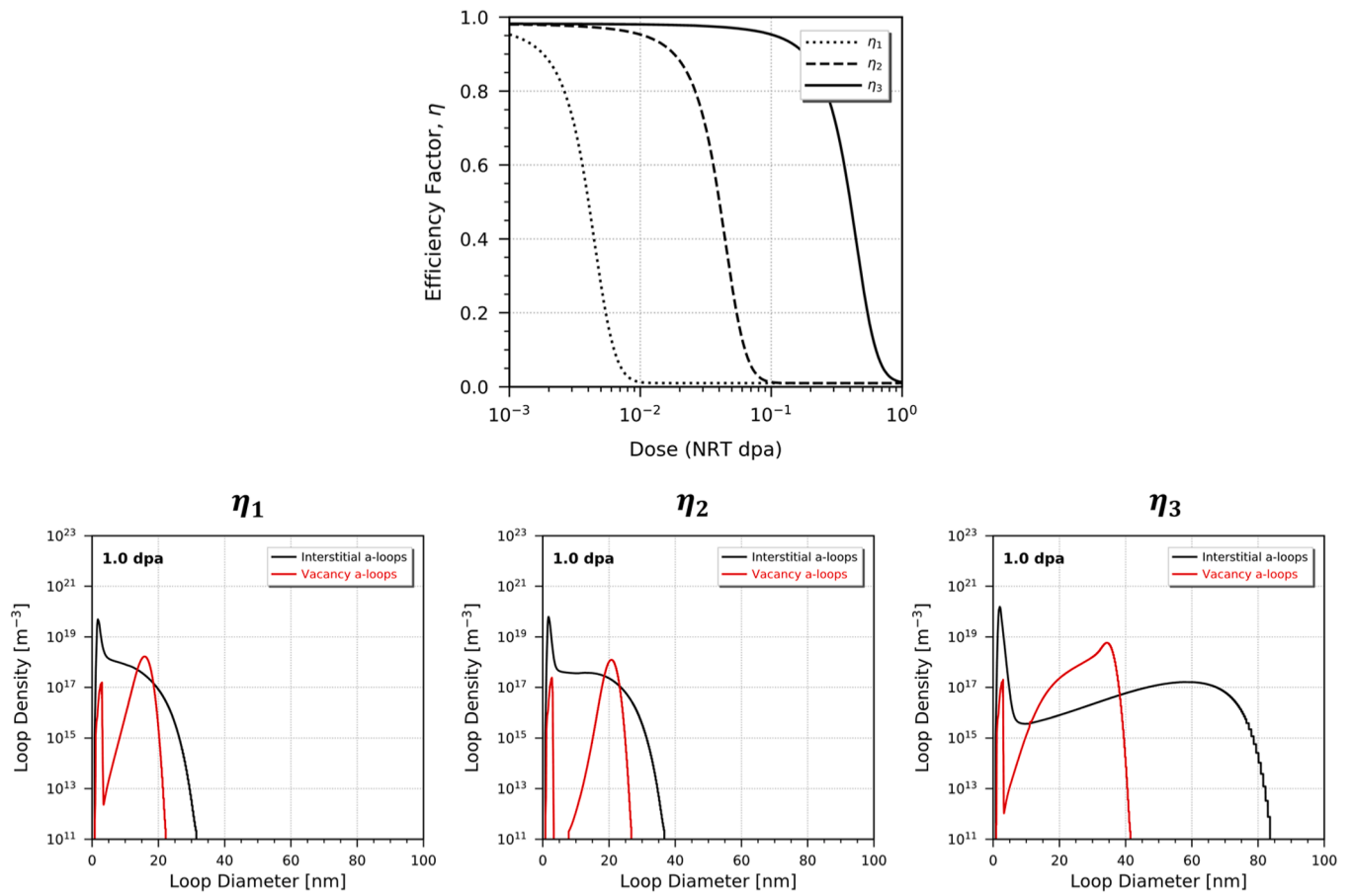


Fig. 14. Three representative curves are shown for the cascade efficiency parameter, η , as a function of dose. The three η curves correspond to saturation of cascade overlap effects at 0.01 dpa (dotted curve), 0.10 dpa (dashed curve), and 1.0 dpa (solid curve) with a residual defect generation rate of 1 %. The resultant vacancy (red) and interstitial (black) a-loop size distributions are shown at 1.0 dpa for simulations using these three curves. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

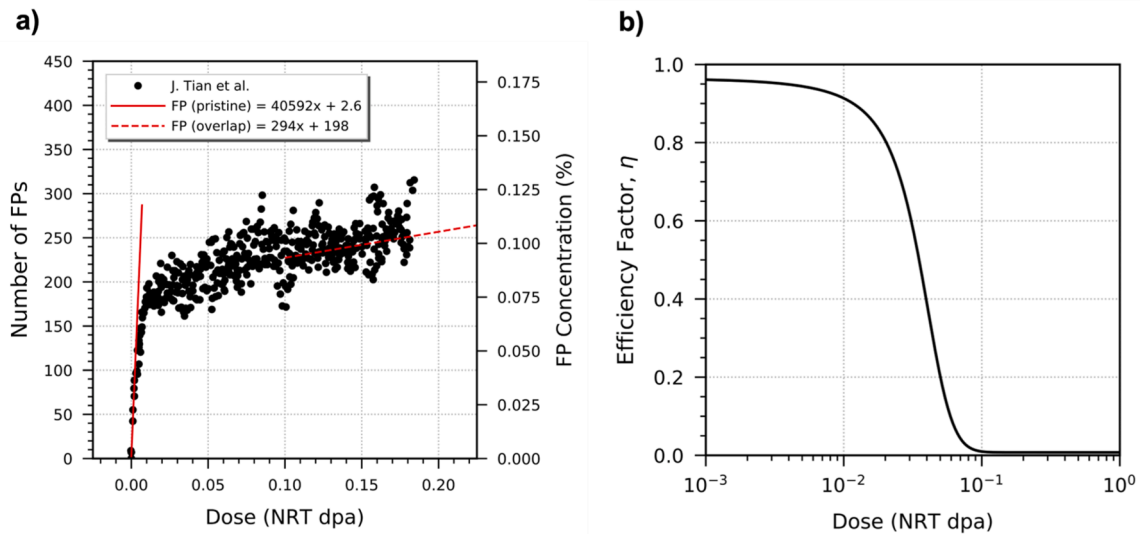


Fig. 15. a) The number of Frenkel pair defects generated by sequential displacement cascades in a single α -zirconium simulation cell (reproduced from the work of J. Tian et al. in Ref. [67] as spherical black markers). Approximate Frenkel pair generation rates are plotted for the pristine and cascade-overlap conditions as solid and dashed red lines, respectively. b) a corresponding cascade efficiency factor curve is plotted based on Eq. (28) using the constants $C_1 = 47$, $C_2 = 1.65$, and $C_3 = 0.0075$. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

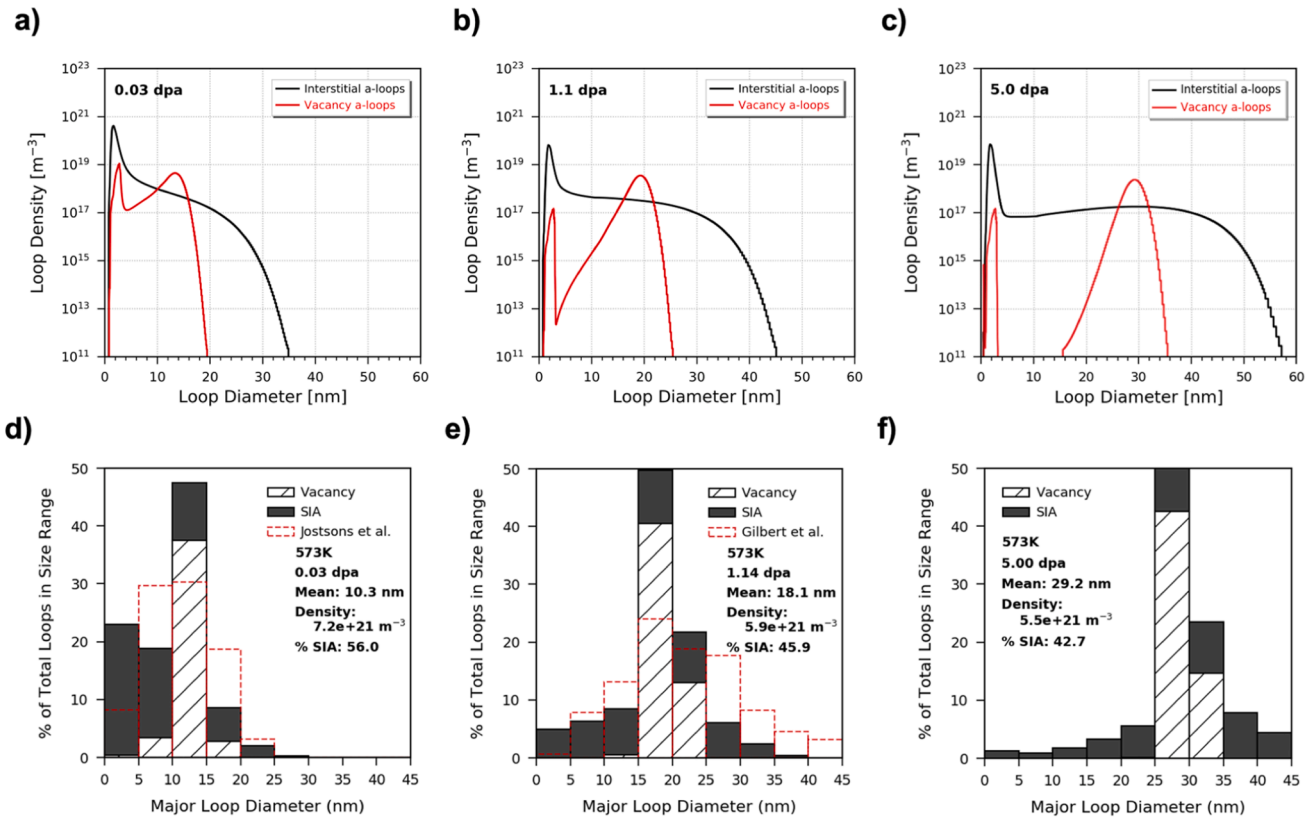


Fig. 16. CD model predictions at $T = 573$ K for the vacancy (red) and interstitial (black) a-loop size distributions at a) 0.033 dpa, b) 1.14 dpa, and c) 5.0 dpa using the full dataset and framework described in this work. Corresponding size distribution histograms are given in d), e), and f), respectively, where vacancy a-loop sizes are represented as dashed white bars. Interstitial a-loop sizes are represented as solid grey bars stacked on top of the vacancy a-loop bars to demonstrate the total combined contributions to the loop size distribution. Experimental loop size histograms from Jostsons et al. [8] and Gilbert et al. [61] are plotted in d) and e), respectively, for comparison. Cascade overlap effects on defect production are incorporated for these simulations based on the cascade efficiency term plotted in Fig. 15(b). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

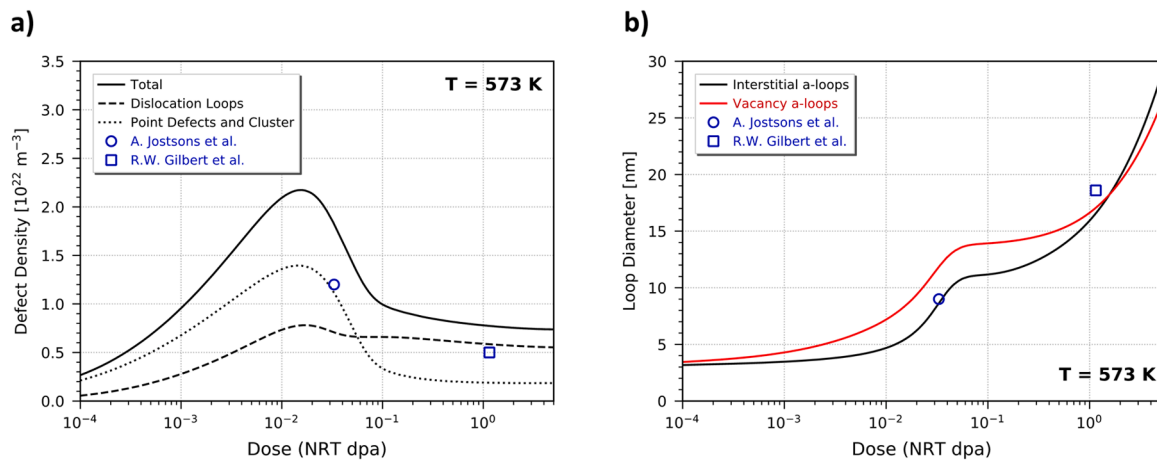


Fig. 17. CD model predictions for a) dislocation loop (dashed line), point defect and small cluster (dotted lines), and total defect (solid line) densities as a function of dpa and b) vacancy (red) and interstitial (black) a-loop diameters as a function of dpa. A-loop characteristics reported by Jostsons et al. [8] and Gilbert et al. [61] are plotted for comparison. Calculations of average loop size and density only consider loops with diameters greater than 3 nm (an approximate experimental resolution limit) to better compare with reported TEM characteristics. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

in a similar fashion as demonstrated here for a-loops. As to the effect of alloying elements, it is clear that the presence of solutes and impurities influences the loop microstructure evolution behavior. Take for example the differences in a-loop characteristics for pure and alloyed zirconium in Table 2. In order to incorporate solute effects into the model, we must

first understand how solutes modify the fundamental mechanisms introduced here. For example, the migration of Fe-V complexes may be facilitated along the c-axis which would alter the diffusion anisotropy ratio of the affected vacancies [70,71]. Similarly, Nb binding with SIAs may restrict their migration and reduce their diffusivities [72,73]. A

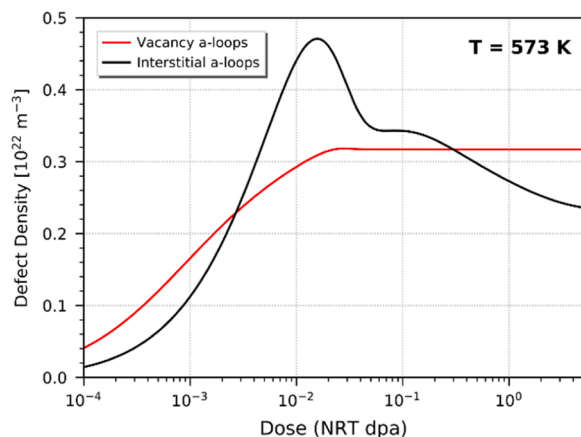


Fig. 18. CD model predictions for vacancy and interstitial a-loop densities as a function of dpa. Calculations of loop densities only consider loops with diameters greater than 3 nm (an approximate experimental resolution limit).

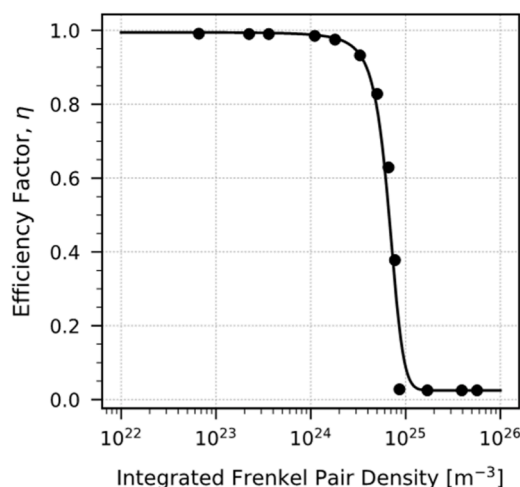


Fig. 19. Proposed cascade efficiency curve (solid line, Eq. (28)) as a function of the integrated Frenkel pair density with $C_1 = 5.3 \times 10^{-25}$, $C_2 = 3.7 \times 10^{24}$, and $C_3 = 0.0075$ for densities in units of m^{-3} . Markers represent the cascade efficiency from Fig. 15(b) plotted against the integrated Frenkel pair density at the corresponding dose.

larger database of lower-length scale data is needed to inform solute-driven mechanisms in a CD approach.

Regarding the spatial dependence of defect clusters, the current 0-D Xolotl CD model assumes that all clusters are randomly and homogeneously distributed. This obviously does not allow for known microstructure behavior such as vacancy a-loop banding [7,8]. Incorporating spatial dependence of vacancy loops may cause self-screening of SIA capture and potentially facilitate the growth of vacancy c-loops within these ordered sheets [29], but future work would be needed to verify this hypothesis. However, considering the current CD model for a-loop nucleation and growth in pure zirconium, the implementation of a wide range of informed physics has facilitated an accurate representation of microstructure evolution in α -zirconium with minimal assumptions.

4. Summary and conclusions

A CD framework is proposed to model microstructure evolution in single-crystalline, pure α -zirconium under neutron irradiation. In order to reduce the number of assumptions typically implemented in microstructure modeling, an extensive database of atomistic simulation data was described and used to inform the underlying mechanisms of loop

growth within the model. Lower-length scale simulations have provided the necessary parameters to model: 1) defect production from displacement cascades (including in-cascade defect recombination and clustering), 2) dissociation rates for all clusters, 3) point defect and defect cluster mobility (including anisotropy), 4) dislocation capture radii with spontaneous and thermal drift methods, and 5) cascade overlap effects on defect generation rates. Defect clusters and small dislocation loops were directly nucleated via a source term in the model. The production rates of point defects are notably lower than the standard NRT rate due to in-cascade recombination and clustering. Implementing dissociation rates for all clusters affects cluster growth and the availability of point defects through continuous monomer emission. Atomistically-informed defect diffusivities affect the rate of cluster interaction; this is particularly true for monovacancies for which the reported diffusivity is considerably higher than was assumed in previous CD models for α -zirconium. Accounting for the anisotropy of defect diffusion modifies the capture efficiencies for dislocation a-loops which varies between interstitial and vacancy a-loops based on their ellipticities. The incorporation of thermal drift point defect capture radii provides an inherent bias for the capture of same-type defects at dislocation loops (i.e. interstitials at interstitial loops and vice versa). Finally, defect production rates were reduced as a function of dpa based on reported behavior for cascade overlap effects.

The resulting atomistically-informed CD model mechanistically predicts the concurrent growth of vacancy and interstitial a-loops in near-equal fractions at 573 K, an aspect of irradiation microstructure that has been challenging to successfully model. The results are consistent with reported a-loop diameters, size distributions, and relative fractions of interstitial-to-vacancy loops for two separate irradiation experiments at low and moderate dose. Vacancy a-loop nucleation is stabilized by their direct production in displacement cascades while vacancy growth is stabilized by the incorporation of thermal capture radii (which favor vacancy capture at vacancy loops). The use of cascade overlap effects on defect generation rates was necessary to predict experimentally consistent evolving a-loop characteristics in zirconium. As such, it is recommended that cascade overlap effects be considered in future mechanistic microstructure evolution models of irradiated zirconium.

CRedit authorship contribution statement

Jose F. March-Rico: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Sophie Blondel:** Writing – review & editing, Supervision, Software, Methodology. **Brian D. Wirth:** Writing – review & editing, Resources, Project administration, Conceptualization.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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